Отчет по Лабораторной работе 7

Выполнила Тарасова Елизавета Андреевна группа 2-МВ-4

1. Вывести список всех процессов системы.

```
elizaveta@ubuntu:~$ ps aux
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root           1  0.0  0.2 168080 11276 ?        Ss   21:02   0:01 /sbin/init a
root           2  0.0  0.0      0     0 ?        S    21:02   0:00 [kthreadd]
root           3  0.0  0.0      0     0 ?        I<   21:02   0:00 [rcu_gp]
root           4  0.0  0.0      0     0 ?        I<   21:02   0:00 [rcu_par_gp]
root           5  0.0  0.0      0     0 ?        I<   21:02   0:00 [slub_flushw
root           6  0.0  0.0      0     0 ?        I<   21:02   0:00 [netns]
root           8  0.0  0.0      0     0 ?        I<   21:02   0:00 [kworker/0:0
root          10  0.0  0.0      0     0 ?        I<   21:02   0:00 [mm_percpu_w
root          11  0.0  0.0      0     0 ?        S    21:02   0:00 [rcu_tasks_r
root          12  0.0  0.0      0     0 ?        S    21:02   0:00 [rcu_tasks_t
root          13  0.0  0.0      0     0 ?        S    21:02   0:00 [ksoftirqd/0
root          14  0.0  0.0      0     0 ?        I    21:02   0:00 [rcu_sched]
root          15  0.0  0.0      0     0 ?        S    21:02   0:00 [migration/0
root          16  0.0  0.0      0     0 ?        S    21:02   0:00 [idle_inject
root          18  0.0  0.0      0     0 ?        S    21:02   0:00 [cpuhp/0]
root          19  0.0  0.0      0     0 ?        S    21:02   0:00 [cpuhp/1]
root          20  0.0  0.0      0     0 ?        S    21:02   0:00 [idle_inject
root          21  0.0  0.0      0     0 ?        S    21:02   0:00 [migration/1
root          22  0.0  0.0      0     0 ?        S    21:02   0:00 [ksoftirqd/1
root          24  0.0  0.0      0     0 ?        I<   21:02   0:00 [kworker/1:0
root          25  0.0  0.0      0     0 ?        S    21:02   0:00 [kdevtmpfs]
root          26  0.0  0.0      0     0 ?        I<   21:02   0:00 [inet_frag_w
root          27  0.0  0.0      0     0 ?        S    21:02   0:00 [kauditd]
root          29  0.0  0.0      0     0 ?        S    21:02   0:00 [khungtaskd]
```

2. Вывести дерево процессов.

Используем pstree

```
elizaveta@ubuntu:~$ pstree
systemd─┬─ModemManager───2*[{ModemManager}]
        ├─NetworkManager───2*[{NetworkManager}]
        ├─accounts-daemon───2*[{accounts-daemon}]
        ├─acpid
        ├─avahi-daemon───avahi-daemon
        ├─colord───2*[{colord}]
        ├─cron
        ├─cups-browsed───2*[{cups-browsed}]
        ├─cupsd
        ├─dbus-daemon
        ├─gdm3─┬─gdm-session-wor─┬─gdm-x-session─┬─Xorg───{Xorg}
        │      │                 │               ├─gnome-session-b─┬─ssh-agent
        │      │                 │               │                 └─2*[{gnome+
        │      │                 │               └─2*[{gdm-x-session}]
        │      │                 └─2*[{gdm-session-wor}]
        │      └─2*[{gdm3}]
        ├─gnome-keyring-d───3*[{gnome-keyring-d}]
        ├─irqbalance───{irqbalance}
        ├─2*[kerneloops]
        ├─networkd-dispat
        ├─polkitd───2*[{polkitd}]
        ├─rsyslogd───3*[{rsyslogd}]
        ├─rtkit-daemon───2*[{rtkit-daemon}]
```

3. С помощью команды top получить список 5 процессов, потребляющих наибольшее количество процессорного времени.

С помощью top получаем список процессов

```
elizaveta@ubuntu:~$ top

top - 22:47:37 up  1:45,  1 user,  load average: 0.02, 0.03, 0.00
Tasks: 270 total,   1 running, 269 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3889.8 total,   2170.6 free,    928.5 used,    790.8 buff/cache
MiB Swap:   1401.6 total,   1401.6 free,      0.0 used.   2731.5 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   2992 elizave+  20   0   20624   3856   3084 R   0.3   0.1   0:00.05 top
      1 root      20   0  168080  11276   8224 S   0.0   0.3   0:01.55 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.01 kthrea+
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_pa+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_f+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworke+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_per+
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     13 root      20   0       0      0      0 S   0.0   0.0   0:00.07 ksofti+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.53 rcu_sc+
     15 root      rt   0       0      0      0 S   0.0   0.0   0:00.04 migrat+
     16 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_i+
     18 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
     19 root      20   0       0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
     20 root     -51   0       0      0      0 S   0.0   0.0   0:00.00 idle_i+
     21 root      rt   0       0      0      0 S   0.0   0.0   0:00.14 migrat+
```

Нажимаем f , дальше выбор TIME+, нажимаем s и q

Далее нажимаем n и указываем колво процессов 5

```
top - 22:48:54 up  1:46,  1 user,  load average: 0.04, 0.04, 0.00
Tasks: 270 total,   1 running, 269 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.8 us,  0.7 sy,  0.0 ni, 98.5 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3889.8 total,   2170.6 free,    928.2 used,    791.1 buff/cache
MiB Swap:   1401.6 total,   1401.6 free,      0.0 used.   2731.7 avail Mem
Maximum tasks = 0, change to (0 is unlimited) 5
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1729 elizave+  20   0 4092548 260548 104532 S   0.7   6.5   0:41.20 gnome-+
   1560 elizave+  20   0  285660  67064  40752 S   1.0   1.7   0:24.02 Xorg
    776 root      20   0  949064  43928  20588 S   0.0   1.1   0:06.65 snapd
   1471 elizave+   9 -11 2015944  20212  15312 S   0.0   0.5   0:04.06 pulsea+
   2610 elizave+  20   0 1205600  73252  46792 S   0.3   1.8   0:03.25 nautil+
   2917 elizave+  20   0  823500  51604  38804 S   1.0   1.3   0:01.57 gnome-+
      1 root      20   0  168080  11276   8224 S   0.0   0.3   0:01.55 systemd
Terminal oot       0 -20       0      0      0 I   0.0   0.0   0:01.36 kworke+
    442 root     -51   0       0      0      0 S   0.0   0.0   0:01.13 irq/16+
   1753 elizave+  20   0  286724  32960  19236 S   0.0   0.8   0:00.86 ibus-e+
    684 systemd+  20   0   90912   5928   5136 S   0.0   0.1   0:00.84 system+
    762 root      20   0  239204  12184   8720 S   0.0   0.3   0:00.80 polkitd
    683 systemd+  20   0   24420  11888   8076 S   0.0   0.3   0:00.77 system+
    730 message+  20   0    9788   6004   3756 S   0.0   0.2   0:00.56 dbus-d+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.54 rcu_sc+
   1481 elizave+  20   0   12872   9932   3804 S   0.0   0.2   0:00.54 dbus-d+
   2862 root      20   0       0      0      0 I   0.0   0.0   0:00.53 kworke+
   1465 elizave+  20   0   21572  12148   8064 S   0.0   0.3   0:00.39 systemd
   1781 elizave+  20   0  162836   7564   6796 S   0.0   0.2   0:00.39 at-spi+
   2129 elizave+  20   0  432284  35920  26228 S   0.0   0.9   0:00.38 update+
    358 root      19  -1   54492  18756  17096 S   0.0   0.5   0:00.37 system+
    318 root      20   0       0      0      0 S   0.0   0.0   0:00.34 jbd2/s+
```

Нажимаем enter и инфа теперь только о 5 процессах

```
top - 22:49:42 up  1:47,  1 user,  load average: 0.02, 0.03, 0.00
Tasks: 270 total,   1 running, 269 sleeping,   0 stopped,   0 zombie
%Cpu(s):  1.0 us,  0.7 sy,  0.0 ni, 98.2 id,  0.2 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3889.8 total,   2170.6 free,    928.1 used,    791.2 buff/cache
MiB Swap:   1401.6 total,   1401.6 free,      0.0 used.   2731.8 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1729 elizave+  20   0 4092552 260536 104532 S   1.0   6.5   0:41.62 gnome-+
   1560 elizave+  20   0  285660  67064  40752 S   1.0   1.7   0:24.41 Xorg
    776 root      20   0  949064  43928  20588 S   1.0   1.1   0:06.70 snapd
   1471 elizave+   9 -11 2015944  20212  15312 S   0.0   0.5   0:04.06 pulsea+
   2610 elizave+  20   0 1205600  73252  46792 S   0.0   1.8   0:03.25 nautil+
```

4. Найти 2 процесса, имеющих более ДВУХ потоков. Использовать состояние процесса

Режим суперпользователя устнавливаем htop : sudo apt install htop

У меня вообще ничего не получилось Я даже с помощью дипсика не могу разобраться как мне подключить сетевое подключение

```
elizaveta@ubuntu:~$ ps -eo pid,ppid,nlwp,comm --sort=-nlwp | awk '$3 > 2'
    PID    PPID NLWP COMMAND
    761       1   11 snapd
   1575    1283    9 evolution-calen
   1292    1283    6 ubuntu-report
   1313    1283    6 gvfsd-fuse
   1509    1283    6 gnome-shell
   1559    1283    6 gnome-shell-cal
   1588    1283    6 evolution-addre
   1658    1495    6 evolution-alarm
    764       1    5 udisksd
   1291    1283    5 tracker-miner-f
   1600    1283    5 gjs
   1670    1283    5 gsd-smartcard
   1818    1283    5 gnome-terminal-
    758       1    4 rsyslogd
   1289    1283    4 pulseaudio
   1295       1    4 gnome-keyring-d
   1332    1283    4 gvfs-udisks2-vo
   1345    1283    4 goa-daemon
   1370    1283    4 gvfs-afc-volume
   1479    1283    4 at-spi-bus-laun
   1495    1283    4 gnome-session-b
   1534    1529    4 ibus-extension-
   1566    1283    4 evolution-sourc
   1619    1283    4 gsd-a11y-settin
   1628    1283    4 gsd-color
   1630    1283    4 gsd-datetime
   1631    1283    4 gsd-housekeepin
```

```
   1840    1283    3 gvfsd-metadata
elizaveta@ubuntu:~$ ps -eo pid,ppid,nlwp,comm --sort=-nlwp | awk '$3 > 2'-$3
awk: line 1: syntax error at or near end of line
elizaveta@ubuntu:~$ htop

Command 'htop' not found, but can be installed with:

sudo apt install htop

elizaveta@ubuntu:~$ sudo apt install htop
[sudo] password for elizaveta:
Reading package lists... Done
Building dependency tree
Reading state information... Done
E: Unable to locate package htop
elizaveta@ubuntu:~$ ping -c 4 8.8.8.8
ping: connect: Network is unreachable
elizaveta@ubuntu:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defau
lt qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: ens33: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc fq_codel state DOW
N group default qlen 1000
    link/ether 00:0c:29:f6:6e:0e brd ff:ff:ff:ff:ff:ff
```

5. Используя команду top, изменить приоритеты 2 процессов.

Top в режиме суперпользователя для возможности понижения приоритета процесса , намимаем r, выбираем процесс с pid1 и меняем параметр на -10

```
elizaveta@ubuntu:~$ sudo top
[sudo] password for elizaveta:

top - 00:14:49 up 42 min,  1 user,  load average: 0.22, 0.08, 0.03
Tasks: 273 total,   1 running, 272 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.3 us,  0.5 sy,  0.0 ni, 99.2 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3889.9 total,   2458.7 free,    822.2 used,    609.0 buff/cache
MiB Swap:   1401.6 total,   1401.6 free,      0.0 used.   2841.0 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1509 elizave+  20   0 3611100 243608 103508 S   1.0   6.1   0:11.38 gnome-+
   1365 elizave+  20   0  281912  65236  40548 S   0.3   1.6   0:07.15 Xorg
   1818 elizave+  20   0  823328  51152  38488 S   0.3   1.3   0:03.73 gnome-+
      1 root      20   0  168296  11456   8268 S   0.0   0.3   0:01.66 systemd
      2 root      20   0       0      0      0 S   0.0   0.0   0:00.02 kthrea+
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_pa+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_f+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworke+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_per+
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
```

Изменения в стобцах PR и NI, для процесса 2 понижаем приоритет

```
    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
   1509 elizave+  20   0 3611100 243596 103508 S   2.1   6.1   0:12.23 gnome-+
   1365 elizave+  20   0  281912  65236  40548 S   1.1   1.6   0:07.73 Xorg
   1818 elizave+  20   0  823328  51152  38488 S   0.7   1.3   0:04.10 gnome-+
   2021 root      20   0   20624   3976   3204 R   0.4   0.1   0:00.63 top
      1 root      10 -10  168296  11456   8268 S   0.0   0.3   0:11.49 systemd
      2 root      30  10       0      0      0 S   0.0   0.0   0:00.02 kthrea+
      3 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_gp
      4 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 rcu_pa+
      5 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 slub_f+
      6 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 netns
      8 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 kworke+
     10 root       0 -20       0      0      0 I   0.0   0.0   0:00.00 mm_per+
     11 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     12 root      20   0       0      0      0 S   0.0   0.0   0:00.00 rcu_ta+
     13 root      20   0       0      0      0 S   0.0   0.0   0:00.04 ksofti+
     14 root      20   0       0      0      0 I   0.0   0.0   0:00.25 rcu_sc+
     15 root      rt   0       0      0      0 S   0.0   0.0   0:00.02 migrat+
```

6. Получить список открытых файлов пользователя aa

Команда lsof ключ –u и имя пользователя

```
elizaveta@ubuntu:~$ lsof -u elizaveta
COMMAND     PID      USER   FD      TYPE             DEVICE SIZE/OFF       NODE
N
s   d    1283 elizaveta  cwd       DIR                8,5     4096          2
/
systemd    1283 elizaveta  rtd       DIR                8,5     4096          2
/
systemd    1283 elizaveta  txt       REG                8,5  1620224    1320002
/usr/lib/systemd/systemd
systemd    1283 elizaveta  mem       REG                8,5  1369384    1321033
/usr/lib/x86_64-linux-gnu/libm-2.31.so
systemd    1283 elizaveta  mem       REG                8,5   178528    1321545
/usr/lib/x86_64-linux-gnu/libudev.so.1.6.17
systemd    1283 elizaveta  mem       REG                8,5  1575112    1321551
/usr/lib/x86_64-linux-gnu/libunistring.so.2.1.0
systemd    1283 elizaveta  mem       REG                8,5   137584    1320763
/usr/lib/x86_64-linux-gnu/libgpg-error.so.0.28.0
systemd    1283 elizaveta  mem       REG                8,5    67912    1320982
/usr/lib/x86_64-linux-gnu/libjson-c.so.4.0.0
systemd    1283 elizaveta  mem       REG                8,5    34872    1320282
/usr/lib/x86_64-linux-gnu/libargon2.so.1
systemd    1283 elizaveta  mem       REG                8,5   431472    1320515
/usr/lib/x86_64-linux-gnu/libdevmapper.so.1.02.1
systemd    1283 elizaveta  mem       REG                8,5    30936    1321578
/usr/lib/x86_64-linux-gnu/libuuid.so.1.3.0
systemd    1283 elizaveta  mem       REG                8,5  2958176    1320462
/usr/lib/x86_64-linux-gnu/libcrypto.so.1.1
systemd    1283 elizaveta  mem       REG                8,5   686160    1321733
```

Files

7. Получить текущее состояние системной памяти

Free выводит информацио о ROM:

```
elizaveta@ubuntu:~$ free
              total        used        free      shared  buff/cache   available
Mem:        3983212      863192     2492504        2068      627516     2887324
Swap:       1435264           0     1435264
elizaveta@ubuntu:~$
```

8. Получить справку об использовании дискового пространства.

Df с ключом –h

```
elizaveta@ubuntu:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            1.9G     0  1.9G   0% /dev
tmpfs           389M  1.8M  388M   1% /run
/dev/sda5        29G  6.7G   21G  25% /
tmpfs           1.9G     0  1.9G   0% /dev/shm
t               5.0M  4.0K  5.0M   1% /run/lock
t               1.9G     0  1.9G   0% /sys/fs/cgroup
/dev/loop0      128K  128K     0 100% /snap/bare/5
/dev/loop1       64M   64M     0 100% /snap/core20/1828
/dev/loop2      347M  347M     0 100% /snap/gnome-3-38-2004/119
/dev/loop3       92M   92M     0 100% /snap/gtk-common-themes/153
/dev/loop4       46M   46M     0 100% /snap/snap-store/638
/dev/loop5       50M   50M     0 100% /snap/snapd/18357
/dev/sda1       511M  4.0K  511M   1% /boot/efi
tmpfs           389M   24K  389M   1% /run/user/1000
/dev/sr0        142M  142M     0 100% /media/elizaveta/CDROM
elizaveta@ubuntu:~$
```

CDROM

9. Вывести информацию о каком-либо процессе, используя содержимое каталога /proc

ls –l , информация о процессе с pid 12 из каталога

```
elizaveta@ubuntu:~$ ls -l /proc/12
ls: cannot read symbolic link '/proc/12/cwd': Permission denied
ls: cannot read symbolic link '/proc/12/root': Permission denied
ls: cannot read symbolic link '/proc/12/exe': Permission denied
total 0
-r--r--r--  1 root root 0 Mar 28 00:29 arch_status
dr-xr-xr-x  2 root root 0 Mar 28 00:29 attr
-rw-r--r--  1 root root 0 Mar 28 00:29 autogroup
-r--------  1 root root 0 Mar 28 00:29 auxv
-r--r--r--  1 root root 0 Mar 28 00:29 cgroup
--w-------  1 root root 0 Mar 28 00:29 clear_refs
-r--r--r--  1 root root 0 Mar 27 23:32 cmdline
-rw-r--r--  1 root root 0 Mar 28 00:29 comm
-rw-r--r--  1 root root 0 Mar 28 00:29 coredump_filter
-r--r--r--  1 root root 0 Mar 28 00:29 cpu_resctrl_groups
-r--r--r--  1 root root 0 Mar 28 00:29 cpuset
lrwxrwxrwx  1 root root 0 Mar 28 00:29 cwd
-r--------  1 root root 0 Mar 28 00:29 environ
lrwxrwxrwx  1 root root 0 Mar 28 00:29 exe
dr-x------  2 root root 0 Mar 28 00:29 fd
dr-xr-xr-x  2 root root 0 Mar 28 00:29 fdinfo
-rw-r--r--  1 root root 0 Mar 28 00:29 gid_map
-r--------  1 root root 0 Mar 28 00:29 io
-r--r--r--  1 root root 0 Mar 28 00:29 limits
-rw-r--r--  1 root root 0 Mar 28 00:29 loginuid
dr-x------  2 root root 0 Mar 28 00:29 map_files
```

10.Вывести информацию о процессоре ПК, используя содержимое каталога /proc

Используем cat

```
elizaveta@ubuntu:~$ cat /proc/cpuinfo
processor       : 0
vendor_id       : GenuineIntel
cpu family      : 6
model           : 165
model name      : Intel(R) Core(TM) i5-10300H CPU @ 2.50GHz
stepping        : 2
microcode       : 0xf0
cpu MHz         : 2496.005
cache size      : 8192 KB
physical id     : 0
siblings        : 1
core id         : 0
cpu cores       : 1
apicid          : 0
initial apicid  : 0
fpu             : yes
fpu_exception   : yes
cpuid level     : 22
wp              : yes
flags           : fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov
clflush mmx fxsr sse sse2 ss syscall nx pdpe1gb rdtscp lm constant_
nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq p
i pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline
```

11.Вывести список модулей, используемых в настоящий момент ядром ОС.

```
elizaveta@ubuntu:~$ cat /proc/modules
isofs 49152 1 - Live 0x0000000000000000
nls_iso8859_1 16384 1 - Live 0x0000000000000000
snd_ens1371 32768 2 - Live 0x0000000000000000
snd_ac97_codec 155648 1 snd_ens1371, Live 0x0000000000000000
binfmt_misc 24576 1 - Live 0x0000000000000000
gameport 24576 1 snd_ens1371, Live 0x0000000000000000
ac97_bus 16384 1 snd_ac97_codec, Live 0x0000000000000000
snd_pcm 135168 2 snd_ens1371,snd_ac97_codec, Live 0x0000000000000000
intel_rapl_msr 20480 0 - Live 0x0000000000000000
intel_rapl_common 40960 1 intel_rapl_msr, Live 0x0000000000000000
snd_seq_midi 20480 0 - Live 0x0000000000000000
snd_seq_midi_event 16384 1 snd_seq_midi, Live 0x0000000000000000
crypto_simd 16384 1 - Live 0x0000000000000000
snd_rawmidi 49152 2 snd_ens1371,snd_seq_midi, Live 0x0000000000000000
ghash_clmulni_intel 16384 0 - Live 0x0000000000000000
aesni_intel 376832 0 - Live 0x0000000000000000
vmw_balloon 24576 0 - Live 0x0000000000000000
crypto_simd 16384 1 aesni_intel, Live 0x0000000000000000
cryptd 24576 2 ghash_clmulni_intel,crypto_simd, Live 0x0000000000000000
rapl 20480 0 - Live 0x0000000000000000
snd_seq 77824 2 snd_seq_midi,snd_seq_midi_event, Live 0x0000000000000000
snd_seq_device 16384 3 snd_seq_midi,snd_rawmidi,snd_seq, Live 0x0000000000000
0
input_leds 16384 0 - Live 0x0000000000000000
```